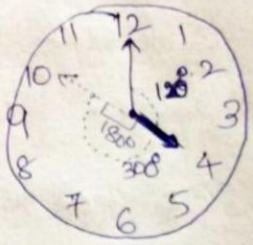
(15)

Note:- B/w 12, 23 ... 1211, hands are maintaining 90° angular difference (or) right angles to each other twice except b/w 23, 34, 89 and 910 [only once in these 4 pair of consecutive hrs]. Total in a day hands are like this 44 times including 3'o clock & 9'o clock [super cases] 2 times in a day.

↳ first coincidance → 12'o clock ] $65\frac{5}{11}$ (or) $1hr\ 5\frac{5}{11}$
2nd " " → $1hr\ 5\frac{5}{11}$ ] $65\frac{5}{11}$ (or) $1hr\ 5\frac{5}{11}$
3rd " → $2hr\ 10\frac{10}{11}$

Note:- for every $65\frac{5}{11}$ minutes hands are coincide to each other.

Ex:-3:- At what time b/w 4'o clock & 5'o clock hands are 180° apart (or) on same st line but in opposite directions?




Sol:- $t = \frac{d}{R.S} = \frac{300^\circ}{5.5} \times \frac{2}{2} = \frac{600}{11}$

$t = 54\frac{6}{11}$ mins

4'o clock + $54\frac{6}{11}$ = 4 hrs $54\frac{6}{11}$ mins

Sol:-2:- $|30H - \frac{11}{2}M| = \theta = 180 \Rightarrow H = 4$

$30 \times 4 - \frac{11}{2}M = 180 \Rightarrow M = -ve$. → Not possible.

$\frac{11}{2}M - 30 \times 4 = 180 \Rightarrow \frac{11}{2}M = 300 \Rightarrow M = \frac{600}{11} = 54\frac{6}{11}$ mins

Sol:-3:- time = $\frac{\text{tra. dist. Min. Spaces}}{11}$

$= 50\frac{50}{11} = 54\frac{6}{11}$

Required time = $4 + 54\frac{6}{11}$ = 4 hrs $54\frac{6}{11}$ mins

Note:- B/w every 2 hr's 1&2, 2&3 ---- 12&1, the hands are 180° apart (or) on same st line but in opposite direction only once except b/w 5&6 and 6&7 [Not possible in these]. Total in a day hands are like this 22 times including 6'o clock [super case] 2 times in a day.

| θ | No. of times in day | No. of times in 12 hr's ($\frac{1}{2}$ day) |
|---|---|---|
| 0°/360° | 22 | 11 |
| 180° | 22 | 11 |
| 90° | 44 | 22 |
| 0<180° | 44 | 22 |
| 180°<360° | 22 | 11 |

↳ Reflex Angular difference:- Angle other than angular difference b/w the hands in clockwise in 360° is Reflex Angle. [360° - A.d b/w the hands].

Ex:- What is reflex Angle b/w the hands @ 4:30?

$A.\theta = \theta = |30H - \frac{11}{2}M|$, $H=4, M=30$

$= |30\times4 - \frac{11}{2}\times30| = |120 - 5.5\times30|$

$= |120-165| = 45^\circ$




Reflex Angular difference $= 360^\circ - A.\theta$

$= 360^\circ - 45^\circ = 315^\circ$

Ex:-5:- How much does a watch lose/gain per day, if hands are coincide for every 63 minutes?

(153)

Ans:- Hands coincide for every $65\frac{5}{11}$ min generally. But in this problem for every 63 minutes.

∴ The loss in $[63<65\frac{5}{11}]$ 63 mins is $65\frac{5}{11}-63 = 2\frac{5}{11}=\frac{27}{11}$ mi

loss in a day or 24 hrs is → ?

63 mins → $\frac{27}{11}$

24hrs = 24X60 → ?

$\Rightarrow \frac{24\times60\times\frac{27}{11}}{63} = \frac{24\times60\times27}{63\times11}$ (3, 7)

Loss = $\frac{4320}{77} = 56\frac{8}{11}$ minutes loss per day.

Note:- If given coincidence $< 65\frac{5}{11}$ ⇒ loss

" " " " $> 65\frac{5}{11}$ ⇒ gain

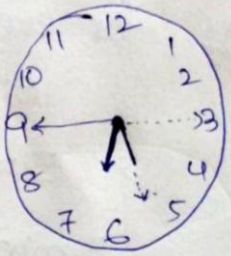
Ex:-6:- What is mirror time of a clock @ 6:45?

Mirror time = By image is 5:15

We have logic for this Q.

Mirror time = 11:60/23:60 - Actual time.

Actual time = 11:60/23:60 - Mirror time.




Ans:- M.T = 11:60 - 6:45 = 5:15

Ex:-7:- A watch is set right at 7am and it gains 5 mins for every 1 hr. What is the time @ 6pm on this watch?

Ans:- gain → Addition, lose → subtraction.

duration of rotation originally = 7am - 6pm = 11 hrs

1hr → 5 min

11hrs → 5X11 = 55 mins gain

So, final time = 6pm + 55 min gaid

= 6:55 pm

Ex:-8:- A watch which gains 5 seconds for every 3 mins watch set @ 7am in the morning. It with this gain watch shows quarter past 4 in the evening of same day, then actual time was?

Ans:-

| | Original | Error |
|---|---|---|
| Initial | 7am | 7am |
| final | ? | 4:15 pm |

] 9hr's 15 mins

Note:- Original Initial & final and Error Initial given asked for Error final then addition/sub according to gain/loss is methodology but here is different scenario

Ans:- for every 3 mins in original clock 3 min 5 sec in error clock rotations. (or)

3 min 5 sec in error clock => 3min in original.

3 min 5 sec ⟶ 3 min.
9hr's 15min ⟶ ?

[Convert all in mins]

$\frac{3\times60+5}{60}$ ⟶ 3 min.
$9\times60+15$ ⟶ ?

$\Rightarrow \frac{555\times3}{185} \Rightarrow \frac{555\times3\times60}{185}$

$\Rightarrow 9\times60$ mins = 9 hr's

Final time in original clock is 7am + 9hr's = 4pm

Ex:-9:- A watch gains uniformly is 5 minutes slow at 6am on monday & is 5 minutes 48 sec fast at 6pm on the following monday. when was it correct?

(155)

Ans:- Time from 6am monday to 6pm on following monday

$= 7$ days & 12 hr's $= 24\times7+12 = 180$ hr's

So, total gain in watch is 5 minutes before 6am & 5min 48 sec after 6pm $\Rightarrow \left(5+5\frac{48}{60}\right) = 10\frac{4}{5} = \frac{54}{5}$ mins.

$\frac{54}{5}$ min gain in 180 hr's. then if that prior 5 mins slow becomes gain then the watch shows right time.

$\frac{54}{5} \longrightarrow 180$ hr's

5min $\longrightarrow$ ?

$\Rightarrow 5\times\frac{180\times5}{54\cdot3} = \frac{250}{3} = 83\frac{1}{3}$ hr's.

$\therefore$ The watch shows correct after $83\frac{1}{3}$ hr's [3 days 11hr's 20 mins] to 6am on monday, which is.

6am Monday + 3days 11hr's 20mins = 5:20 pm Thursday

Ex:-10:- In a watch the bell rings @ 1'o clock once, 2'o clock twice, 3'o clock thrice ..... and at 6'o clock the bell takes 30 seconds to ring for 6 times then. how much time it takes to ring @ 10'o clock?

Ans:- At 6'o clock. 6 rings like.

|t•|t•|t•|t•|t•|

30 seconds

5 times of time gaps = 30 sec $\Rightarrow t = \frac{30}{5} = 6$ sec's

At 10'o clocks 10 rings means 9 time gaps.

So, $9\times6 = 54$ sec's bell takes to ring for 10 times.